Alexander S. Rose and Peter W. Hildebrand. NGL Viewer: a web application for molecular visualization. *Nucleic Acids Research*, 43(W1):W576–W579, 04 2015. ISSN 0305-1048. doi: 10.1093/nar/gkv402. URL https://doi.org/10.1093/nar/gkv402.

Alexander S Rose, Anthony R Bradley, Yana Valasatava, Jose M Duarte, Andreas Prlić, and Peter W Rose. NGL viewer: web-based molecular graphics for large complexes. *Bioinformatics*, 34(21): 3755–3758, 05 2018. ISSN 1367-4803. doi: 10.1093/bioinformatics/bty419. URL https://doi.org/10.1093/bioinformatics/bty419.

Charles Rubin and Ora Rosen. Protein phosphorylation. *Annual Review of Biochemistry*, 44:831–887, 1975. URL https://doi.org/10.1146/annurev.bi.44.070175.004151.

Philippe Schwaller, Benjamin Hoover, Jean-Louis Reymond, Hendrik Strobelt, and Teodoro Laino. Unsupervised attention-guided atom-mapping. *ChemRxiv*, 5 2020. doi: 10.26434/chemrxiv.12298559.v1. URL https://chemrxiv.org/articles/Unsupervised_Attention-Guided_Atom-Mapping/12298559.

Sofia Serrano and Noah A. Smith. Is attention interpretable? In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, pp. 2931–2951, Florence, Italy, July 2019. Association for Computational Linguistics. doi: 10.18653/v1/P19-1282. URL https://www.aclweb.org/anthology/P19-1282.

Martin Steinegger and Johannes Söding. Clustering huge protein sequence sets in linear time. *Nature Communications*, 9(2542), 2018. doi: 10.1038/s41467-018-04964-5.

Baris E. Suzek, Yuqi Wang, Hongzhan Huang, Peter B. McGarvey, Cathy H. Wu, and the UniProt Consortium. UniRef clusters: a comprehensive and scalable alternative for improving sequence similarity searches. *Bioinformatics*, 31(6):926–932, 11 2014. ISSN 1367-4803. doi: 10.1093/bioinformatics/btu739. URL https://doi.org/10.1093/bioinformatics/btu739.

Yi Chern Tan and L. Elisa Celis. Assessing social and intersectional biases in contextualized word representations. In *Advances in Neural Information Processing Systems 32*, pp. 13230–13241. Curran Associates, Inc., 2019.

Ian Tenney, Dipanjan Das, and Ellie Pavlick. BERT rediscovers the classical NLP pipeline. In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, pp. 4593–4601, Florence, Italy, 2019. Association for Computational Linguistics.

Shikhar Vashishth, Shyam Upadhyay, Gaurav Singh Tomar, and Manaal Faruqui. Attention interpretability across NLP tasks. *arXiv preprint arXiv:1909.11218*, 2019.

Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. In *Advances in Neural Information Processing Systems*, pp. 5998–6008, 2017.

Sara Veldhoen, Dieuwke Hupkes, and Willem H. Zuidema. Diagnostic classifiers revealing how neural networks process hierarchical structure. In *CoCo@NIPS*, 2016.

Jesse Vig. A multiscale visualization of attention in the Transformer model. In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics: System Demonstrations*, pp. 37–42, Florence, Italy, 2019. Association for Computational Linguistics.

Jesse Vig and Yonatan Belinkov. Analyzing the structure of attention in a Transformer language model. In *Proceedings of the 2019 ACL Workshop BlackboxNLP: Analyzing and Interpreting Neural Networks for NLP*, pp. 63–76, Florence, Italy, 2019. Association for Computational Linguistics.

Jesse Vig, Sebastian Gehrmann, Yonatan Belinkov, Sharon Qian, Daniel Nevo, Yaron Singer, and Stuart Shieber. Investigating gender bias in language models using causal mediation analysis. In *Advances in Neural Information Processing Systems*, volume 33, pp. 12388–12401, 2020.

Gregor Wiedemann, Steffen Remus, Avi Chawla, and Chris Biemann. Does BERT make any sense? Interpretable word sense disambiguation with contextualized embeddings. *arXiv preprint arXiv:1909.10430*, 2019.

Sarah Wiegreffe and Yuval Pinter. Attention is not not explanation. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, pp. 11–20, November 2019.

Zhilin Yang, Zihang Dai, Yiming Yang, Jaime Carbonell, Russ R Salakhutdinov, and Quoc V Le. XLNet: Generalized autoregressive pretraining for language understanding. In H. Wallach, H. Larochelle, A. Beygelzimer, F. d'Alché-Buc, E. Fox, and R. Garnett (eds.), *Advances in Neural Information Processing Systems*, volume 32, pp. 5753–5763. Curran Associates, Inc., 2019.

Ruiqi Zhong, Steven Shao, and Kathleen McKeown. Fine-grained sentiment analysis with faithful attention. *arXiv preprint arXiv:1908.06870*, 2019.

A MODEL OVERVIEW

A.1 PRE-TRAINED MODELS

Table 1 provides an overview of the five pre-trained Transformer models studied in this work. The models originate from the TAPE and ProtTrans repositories, spanning three model architectures: BERT, ALBERT, and XLNet.

Table 1: Summary of pre-trained models analyzed, including the source of the model, the type of Transformer used, the number of layers and heads, the total number of model parameters, the source of the pre-training dataset, and the number of protein sequences in the pre-training dataset.

Source	Name	Type	Layers	Heads	Params	Train Dataset	# Seq
TAPE	TapeBert	BERT	12	12	94M	Pfam	31M
ProtTrans	ProtBert	BERT	30	16	420M	Uniref100	216M
ProtTrans	ProtBert-BFD	BERT	30	16	420M	BFD	2.1B
ProtTrans	ProtAlbert	ALBERT	12	64	224M	Uniref100	216M
ProtTrans	ProtXLNet	XLNet	30	16	409M	Uniref100	216M

A.2 BERT TRANSFORMER ARCHITECTURE

Stacked Encoder: BERT uses a stacked-encoder architecture, which inputs a sequence of tokens $\boldsymbol{x} = (x_1, ..., x_n)$ and applies position and token embeddings followed by a series of encoder layers. Each layer applies multi-head self-attention (see below) in combination with a feedforward network, layer normalization, and residual connections. The output of each layer ℓ is a sequence of contextualized embeddings $(\mathbf{h}_1^{(\ell)}, \ldots, \mathbf{h}_n^{(\ell)})$.

Self-Attention: Given an input $\boldsymbol{x} = (x_1, \ldots, x_n)$, the self-attention mechanism assigns to each token pair i, j an attention weight $\alpha_{i,j} > 0$ where $\sum_j \alpha_{i,j} = 1$. Attention in BERT is bidirectional. In the multi-layer, multi-head setting, α is specific to a layer and head. The BERT-Base model has 12 layers and 12 heads. Each attention head learns a distinct set of weights, resulting in 12 x 12 = 144 distinct attention mechanisms in this case.

The attention weights $\alpha_{i,j}$ are computed from the scaled dot-product of the *query vector* of i and the *key vector* of j, followed by a softmax operation. The attention weights are then used to produce a weighted sum of value vectors:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \tag{2}$$

using query matrix Q, key matrix K, and value matrix V, where d_k is the dimension of K. In a multi-head setting, the queries, keys, and values are linearly projected h times, and the attention operation is performed in parallel for each representation, with the results concatenated.

A.3 OTHER TRANSFORMER VARIANTS

ALBERT: The architecture of ALBERT differs from BERT in two ways: (1) It shares parameters across layers, unlike BERT which learns distinct parameters for every layer and (2) It uses factorized embeddings, which allows the input token embeddings to be of a different (smaller) size than the hidden states. The original version of ALBERT designed for text also employed a sentence-order prediction pretraining task, but this was not used on the models studied in this paper.

XLNet: Instead of the masked-language modeling pretraining objective use for BERT, XLNet uses a bidirectional auto-regressive pretraining method that considers all possible orderings of the input factorization. The architecture also adds a segment recurrence mechanism to process long sequences, as well as a relative rather than absolute encoding scheme.